

D671

Elementary Science Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

Third graders chart the time of sunset every week for four months on a bulletin board decorated with stars, then notice that sunset gets earlier each week in fall. The class uses a new weather station. Which crosscutting concept are they using?

- A. Analyzing and interpreting data
- B. Weather and climate
- C. Structure and function
- D. Patterns

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A class places a sealed bag with 50 g of ice on a scale. The scale is new and the classroom is 22°C. After the ice melts, what will the scale read?

- A. More than 50 g, because liquid spreads out.
- B. It depends on the room temperature.
- C. 50 g
- D. Less than 50 g, because water is lighter than ice.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A student stirs 10 g of salt into 200 g of water until it disappears. Total mass afterward is 210 g. What does this show?

- A. The salt is still there, dissolved; matter was conserved.
- B. A chemical reaction created new mass.
- C. The salt was destroyed when it dissolved.
- D. The scale must be wrong, since the salt disappeared.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a meadow food chain, grass captures 10,000 units of energy. Rabbits eat the grass, and foxes eat the rabbits. The meadow is near a stream, and hawks nest nearby. Using the 10% rule, how much energy reaches the foxes?

- A. 100 units
- B. 10,000 units
- C. 1,000 units
- D. 9,000 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Owls in a food chain receive 50 units of energy from mice. Using the 10% rule, how much energy did the plants the mice ate originally capture?

- A. 500 units
- B. 50 units
- C. 5,000 units
- D. 5 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A willow tree was planted in 90 kg of soil. Five years later the tree gained about 74 kg, while the soil lost only about 0.06 kg. The pot was blue and was kept by a sunny window. Where did most of the tree's added mass come from?

- A. Sunlight turning into matter.
- B. The blue color of the pot trapping heat.
- C. Carbon dioxide from the air and water, combined through photosynthesis.
- D. The soil.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student says it is summer in the Northern Hemisphere because Earth is closest to the sun then. She loves astronomy and has a telescope at home. Which fact best challenges her idea?

- A. Earth is always the same distance from the sun.
- B. Her telescope is not strong enough to see it.
- C. The sun is hotter in summer.
- D. When it is summer in the Northern Hemisphere, it is winter in the Southern Hemisphere, though both are the same distance from the sun.

Target word _____	The one fact that decides _____	Answer __	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

If Earth's axis had no tilt, what would happen to seasons?

- A. Seasons would be stronger.
- B. Every day would be summer everywhere.
- C. Nothing would change, since distance causes seasons.
- D. Seasons would mostly disappear, since sunlight angle and day length would stay nearly the same year-round.

Target word _____	The one fact that decides _____	Answer __	[] sure [] unsure [] guess
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9.

Water seeps into cracks in a boulder, freezes, and splits the rock over many winters. The boulder sits in a national park that gets 2 million visitors a year. Which process is this?

- A. Human impact from visitors
- B. Weathering
- C. Erosion
- D. Deposition

Target word _____	The one fact that decides _____	Answer __	[] sure [] unsure [] guess
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10.

A toy car rolls across a smooth floor at a steady speed in a straight line. It is red, has four rubber wheels, and was a birthday gift. What is true about the forces on it?

- A. They are balanced.
- B. They are unbalanced, because it is moving.
- C. They depend on the car's color.
- D. There are no forces on it.

Target word _____	The one fact that decides _____	Answer __	[] sure [] unsure [] guess
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11. [TRANSFER — reversed stem / different variable]

A soccer ball at rest suddenly begins to move and speeds up. What must be true?

- A.** The forces were balanced.
- B.** An unbalanced force acted on it.
- C.** No force acted on it.
- D.** Gravity alone pushed it forward.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A student says the moon looks like a crescent because Earth's shadow covers part of it. The class watches the moon from a school field with a large oak tree. What is the correct explanation?

- A.** Earth's shadow covers part of the moon.
- B.** As the moon orbits Earth, we see different amounts of its sunlit half.
- C.** Trees or clouds block part of the moon.
- D.** The moon makes less light at a crescent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

Third graders chart the time of sunset every week for four months ~~on a bulletin board decorated with stars,~~ then notice that sunset gets earlier each week in fall. ~~The class uses a new weather station.~~ Which crosscutting concept are they using?

Target word: identify the crosscutting concept **Decides:** Noticing a regular change in data.

A	Confusion set	That is a science practice, not a crosscutting concept.
B	Camouflage	A content topic suggested by the weather station, not a crosscutting concept.
C	Adjacent category	Relates shape to purpose, not a data trend.
D	Correct	Students identify a pattern in observations over time.

2. Answer C Concept 2

A class places a sealed bag with 50 g of ice on a scale. ~~The scale is new and the classroom is 22°C.~~ After the ice melts, what will the scale read?

Target word: mass after melting **Decides:** Physical change in a closed bag conserves mass.

A	Inverted operation	Volume change is not mass change.
B	Camouflage	Temperature speeds melting but does not change mass.
C	Correct	Mass is conserved; only the state changed.
D	Confusion set	Confuses density with mass.

3. Answer A Concept 2 · transfer

A student stirs 10 g of salt into 200 g of water until it disappears. Total mass afterward is 210 g. What does this show?

Target word: reverse: explain a result **Decides:** Mass conserved though salt is not visible.

A	Correct	$10 + 200 = 210$ g.
B	Adjacent category	Dissolving is physical, and mass is never created.
C	Inverted operation	Mass would drop to 200 g.
D	Absolute language	Not seeing a substance does not mean it is gone.

4. Answer A Concept 3

In a meadow food chain, grass captures 10,000 units of energy. Rabbits eat the grass, and foxes eat the rabbits. ~~The meadow is near a stream, and hawks nest nearby.~~ Using the 10% rule, how much energy reaches the foxes?

Target word: energy at third level **Decides:** $10,000 \rightarrow 1,000 \rightarrow 100$.

A	Correct	10% of 10,000 = 1,000 to rabbits; 10% of that = 100 to foxes.
B	Absolute language	Assumes all energy passes up the chain.
C	Dropped element	Stops at the rabbits.
D	Inverted operation	That is the energy not passed to rabbits.

5. Answer C Concept 3 • transfer

Owls in a food chain receive 50 units of energy from mice. Using the 10% rule, how much energy did the plants the mice ate originally capture?

Target word: solve backward for the producer **Decides:** $50 \times 10 \times 10$.

A	Dropped element	Goes back only one level.
B	Wrong stem number	Repeats the owl's energy.
C	Correct	Mice had 500; plants had 5,000.
D	Inverted operation	Divides instead of multiplies going backward.

6. Answer C Concept 4

A willow tree was planted in 90 kg of soil. Five years later the tree gained about 74 kg, while the soil lost only about 0.06 kg. ~~The pot was blue and was kept by a sunny window.~~ Where did most of the tree's added mass come from?

Target word: source of plant mass **Decides:** Soil barely changed; mass came from air and water.

A	Adjacent category	Light supplies energy, not mass.
B	Camouflage	Pot color is irrelevant.
C	Correct	Photosynthesis builds sugars from CO ₂ and water.
D	Confusion set	The soil lost only a tiny amount.

7. Answer D Concept 5

A student says it is summer in the Northern Hemisphere because Earth is closest to the sun then. ~~She loves astronomy and has a telescope at home.~~ Which fact best challenges her idea?

Target word: cause of seasons **Decides:** Hemispheres have opposite seasons at the same distance.

A	Absolute language	The orbit is slightly elliptical; distance just is not the cause.
B	Camouflage	Irrelevant to the cause of seasons.
C	Adjacent category	The sun's output does not change seasonally.
D	Correct	Distance cannot explain opposite seasons; tilt can.

8. Answer D Concept 5 • transfer

If Earth's axis had no tilt, what would happen to seasons?

Target word: reverse: remove the cause **Decides:** Tilt causes seasons.

A	Inverted operation	Removing tilt weakens seasons.
B	Absolute language	Climate would vary by latitude, not be summer everywhere.
C	Confusion set	Repeats the misconception.
D	Correct	Without tilt, the main cause is gone.

9. Answer B Concept 6

Water seeps into cracks in a boulder, freezes, and splits the rock over many winters. The boulder sits in a national park ~~that gets 2 million visitors a year~~. Which process is this?

Target word: weathering vs. erosion **Decides:** Rock broken in place, not moved.

A	Camouflage	The stem describes ice, not people.
B	Correct	Freeze-thaw breaks rock where it is.
C	Confusion set	Erosion moves material.
D	Adjacent category	Deposition drops moved material.

10. Answer A Concept 7

A toy car rolls across a smooth floor at a steady speed in a straight line. ~~It is red, has four rubber wheels, and was a birthday gift.~~ What is true about the forces on it?

Target word: forces at constant velocity **Decides:** No change in motion.

A	Correct	Steady speed in a straight line means net force is zero.
B	Confusion set	Motion itself does not require net force.
C	Camouflage	Color is irrelevant.
D	Absolute language	Gravity, the floor, and friction still act.

11. Answer B Concept 7 · transfer

A soccer ball at rest suddenly begins to move and speeds up. What must be true?

Target word: reverse: from motion change to force **Decides:** Motion changed.

A	Inverted operation	Balanced forces produce no change.
B	Correct	Changes in motion require a net force.
C	Dropped element	Something had to start it moving.
D	Adjacent category	Gravity pulls down, not forward.

12. Answer B Concept 8

A student says the moon looks like a crescent because Earth's shadow covers part of it. ~~The class watches the moon from a school field with a large oak tree.~~ What is the correct explanation?

Target word: cause of phases **Decides:** We see varying portions of the lit half.

A	Confusion set	That describes a lunar eclipse.
B	Correct	Phases come from viewing angle.
C	Camouflage	Local objects do not cause phases.
D	Adjacent category	The moon reflects sunlight; it does not make light.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Crosscutting concepts	Crosscutting concepts (patterns; cause and effect; scale; systems; energy and matter; structure and function; stability and change) are lenses that connect ideas across all science disciplines.	Confusing a crosscutting concept with a science practice or a content topic.	A lesson goal framed as a way of thinking across topics.
2	Conservation of matter	In physical changes and in closed-system chemical reactions, total mass stays the same; matter changes form but is not created or destroyed.	Thinking dissolved or melted substances weigh less or disappear.	Mass before and after melting, dissolving, or mixing.
3	Energy flow in food chains	Energy enters from the sun through producers, and only about 10% passes to each next level; the rest is used or lost as heat.	Believing energy is recycled like matter or that all energy passes up the chain.	A food chain with energy amounts.
4	Where plant mass comes from	Plants build most of their mass from carbon dioxide in the air and water, using light energy in photosynthesis; soil supplies small amounts of minerals.	Believing plants get their food or mass from the soil.	A plant grows much heavier while soil mass barely changes.
5	Cause of seasons	Seasons are caused by Earth's tilted axis, which changes how directly sunlight strikes and how long days are; not by distance from the sun.	Explaining summer as Earth being closer to the sun.	Opposite seasons in the two hemispheres.
6	Weathering, erosion, deposition	Weathering breaks rock down in place; erosion moves the pieces; deposition drops them in a new place.	Using weathering and erosion as synonyms.	Whether rock is broken, moved, or dropped.
7	Balanced and unbalanced forces	Balanced forces cause no change in motion (an object stays still or keeps a steady speed); unbalanced forces change speed or direction.	Thinking a moving object must have an unbalanced force on it.	An object at rest or moving at constant speed.
8	Moon phases	Phases come from how much of the moon's sunlit half we see as it orbits Earth; Earth's shadow causes eclipses, not phases.	Explaining phases with Earth's shadow.	A question about why the moon's shape appears to change.

Course scope (public description): Elementary Science Curriculum is a comprehensive course that provides an in-depth look at the development and implementation of a robust, inquiry-based science curriculum that aligns with current educational standards and best practices. Participants explore key concepts in teaching science to young learners, including major concepts within the Earth, life, and physical sciences. The curriculum emphasizes the importance of cross-cutting concepts, science and engineering practices, and disciplinary core ideas in elementary science education to support students' conceptual understanding. By the end of the course, educators are prepared with the foundational knowledge and tools necessary to inspire a lifelong interest in science among their students.